Machine learning take-home assignment 2

Binary classification task
social media accounts

Course: Introduction to Machine Learning (822047-B-6) Spring 2025

Çiçek Güven

Görkem Saygılı

Student: Aiden Ricksen

Student number: 2117510

Introduction

This report details a binary classification task focused on distinguishing between automated and genuine user accounts on a social media platform. The dataset includes both numerical and categorical features representing user behavior and profile characteristics. The classification objective was addressed through a supervised learning approach, employing a Random Forest and Logistic Regression model. The performance of both models was evaluated using a consistent pipeline encompassing preprocessing, hyperparameter tuning, and final testing

Models

Logistic Regression was used as a baseline classifier because it provides a stable and interpretable framework for binary classification problems. It estimates the probability of class membership using a logistic function and performs well when the data is linearly separable. Furthermore, its simplicity makes it an appropriate reference point for evaluating more complex models. Regularization was incorporated through an L2 penalty to prevent overfitting and control model complexity.

The Random Forest Classifier was selected as a more expressive model capable of modeling non-linear relationships. It builds multiple decision trees and combines their outputs, which typically results in improved generalization and more accurate prediction. Random Forests are also less sensitive to outliers and irrelevant features, and they provide inherent measures of feature importance that can aid interpretability. This model was particularly suitable for the mixed-type dataset, which included both continuous variables such as follower count and discrete variables such as profile photo presence.

Both models are expected to generalize well to similar datasets, provided the structure of the input features remains consistent. The preprocessing pipeline was designed to automatically handle missing values and encode categorical variables, allowing the same codebase to be reused on similar datasets without manual intervention. Logistic Regression is particularly effective on smaller datasets due to its simplicity and low variance, while Random Forest can leverage larger datasets more effectively through its ensemble structure, offering improved generalization in the presence of more data and feature diversity

Data Preparation

The dataset was composed of user profile features, including activity metrics and categorical indicators. Missing data in numerical features were represented as NaN, while categorical features included the label “Unknown” for missing values. A preprocessing pipeline was constructed to handle these characteristics systematically. Numerical features were imputed using the median strategy, which is robust to outliers, and were then standardized to ensure uniform scaling. Missing-value indicators were also appended to preserve information about missingness. For categorical features, missing values were imputed with a constant “Unknown” category before applying one-hot encoding. This approach allowed categorical variables to be represented without introducing assumptions about ordinal relationships.

To enable fair and reproducible model evaluation, the dataset was split using a stratified three-way division. Twenty percent of the data was held out as a test set using a fixed random seed for reproducibility. From the remaining data, twenty percent was further reserved for validation, resulting in 64% of the data for training, 16% for validation, and 20% for testing. Stratification ensured that class distributions were preserved across all partitions, which is critical for evaluating classification models fairly.

Model Training

Hyperparameter tuning was conducted using grid search with the F1-score on the validation set as the evaluation metric. For Logistic Regression, the primary hyperparameter was the regularization strength C, tested across the values {0.01, 0.1, 1, 10, 100, 1000}. The best result was achieved at C = 10, which balanced regularization and model flexibility. Too low a value of C (strong regularization) led to underfitting, while very high values resulted in overfitting to the training data. The solver selected was liblinear, which is well-suited for small to medium-sized datasets and supports both L1 and L2 regularization. It is based on a coordinate descent algorithm, which ensures fast and stable convergence for binary classification tasks. While alternative solvers such as saga and lbfgs are available, they are typically optimized for large-scale or multinomial problems and did not offer an advantage in this setting and similar datasets.

For the Random Forest Classifier, a broader grid was explored to optimize ensemble complexity and generalization. The number of trees was tuned at 100 and 200, while the maximum tree depth was varied among None, 10, and 20. The minimum number of samples per leaf was set to 1 or 5. The optimal configuration was found to be 100 estimators, unlimited tree depth (None), and a minimum leaf size of 1. These settings allowed the model to fully exploit the available feature interactions without prematurely pruning informative branches. While deeper trees and smaller leaf sizes increased model complexity, the validation results confirmed that this configuration yielded strong generalization without overfitting.

Results and Discussion

Both models demonstrated high levels of accuracy and generalization on the test set. Logistic Regression achieved an accuracy of 0.969, a precision of 0.977, a recall of 0.959, an F1-score of 0.968, and an AUC of 0.994. The Random Forest model matched this accuracy at 0.969, exceeded Logistic Regression in precision with 0.984, slightly underperformed in recall with 0.954, delivered an F1-score of 0.969, and achieved an AUC of 0.987 (Figure 1). The ROC and Precision-Recall (PR) curves further illustrate the models' classification performance. As shown in Figure 3, both the Logistic Regression and Random Forest classifiers exhibit ROC curves that closely approach the top-left corner, reflecting a strong trade-off between true positive rate and false positive rate. The area under the ROC curve (AUC) is notably high for both models—0.994 for Logistic Regression and 0.987 for Random Forest—indicating strong overall discriminative ability. Figure 4 depicts the precision-recall curves, which are particularly informative in the context of class imbalance, as they focus solely on performance with respect to the positive class (i.e., bots). Both curves remain close to the upper-right region, confirming that high

precision is maintained even as recall increases. Logistic Regression displays slightly smoother performance across thresholds due to its probabilistic nature, while Random Forest exhibits marginally higher precision at the upper end. These visualizations reinforce the numerical findings and confirm that both models effectively distinguish bots from genuine users across decision thresholds.

This conclusion is further supported by the confusion matrices shown in Figure 2. The majority of predictions were correct for both models, with relatively few false positives or false negatives. Logistic Regression yielded slightly fewer false negatives, consistent with its higher recall, while Random Forest achieved fewer false positives, aligning with its superior precision. These matrices highlight the models' reliability: genuine users were rarely misclassified as bots, and most bot accounts were accurately identified. The strong diagonal dominance in the matrices visually confirms the effectiveness of both classifiers and further validates the high performance reported through standard evaluation metrics.

Group-level performance was also evaluated across gender categories, with metrics presented in Figure 8. For male users, the model achieved high precision and recall, resulting in an F1-score of 0.973. Female users also experienced high performance, with an F1-score of 0.957. However, categories labeled “Unknown” and “Other” contained too few samples (2 and 7, respectively) to support reliable evaluation. This disparity underscores the importance of sufficient representation in fairness assessments.

To assess model interpretability, SHAP values were used to identify influential features in the Random Forest model. The top predictive features included follower count, status count, and the presence of a profile photo. These features were consistently ranked as having high importance (Figure 5), aligning with expectations that bots tend to exhibit distinctive behavioral signatures, such as abnormal activity levels or incomplete profiles.

Conclusion

The classification system developed demonstrates strong predictive accuracy and generalization. Logistic Regression, despite its simplicity, matched the performance of the more complex Random Forest model, suggesting that the problem space is largely amenable to linear modeling. Preprocessing techniques such as imputation, standardization, and one-hot encoding were critical in ensuring data quality and compatibility with both models. The consistent performance across demographic groups and the interpretability of feature importances reinforce the stability of the approach. Within the scope of this project, future improvements could include more robust subgroup evaluation, simple feature transformations to enhance separability, and calibrated probability outputs to better support real-world decision-making thresholds. Additionally, implementing stratified cross-validation would improve the generalizability and reusability of the solution.

Appendix

Figure 1:

Model	Accuracy	Precision	Recall	F1-score	AUC
Logistic regression	0.969	0.977	0.959	0.968	0.994
Random Forest Classifier	0.969	0.984	0.954	0.969	0.987

Figure 2:

a: b:

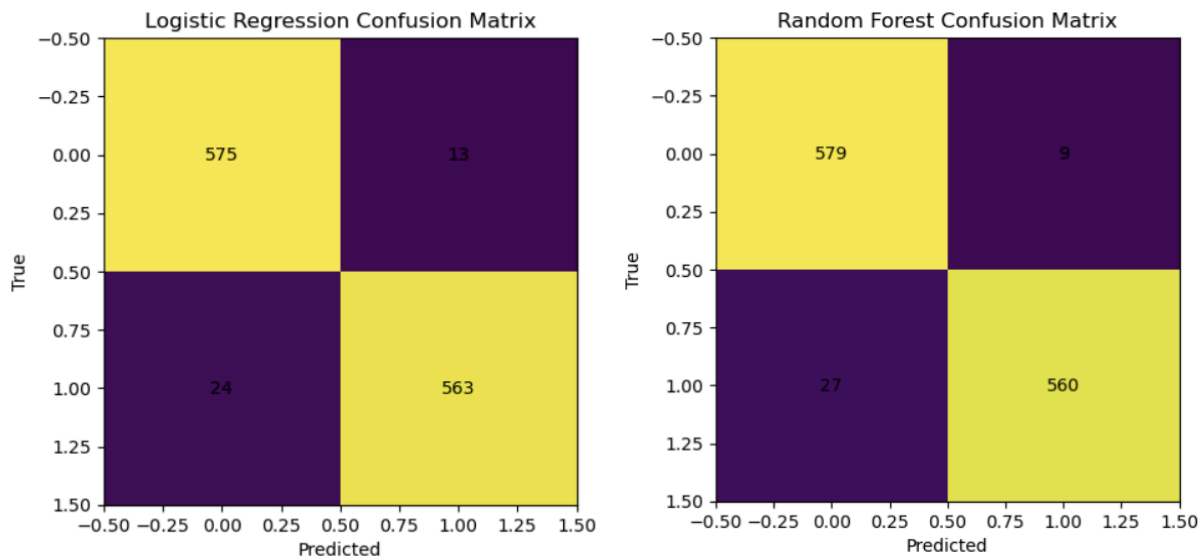
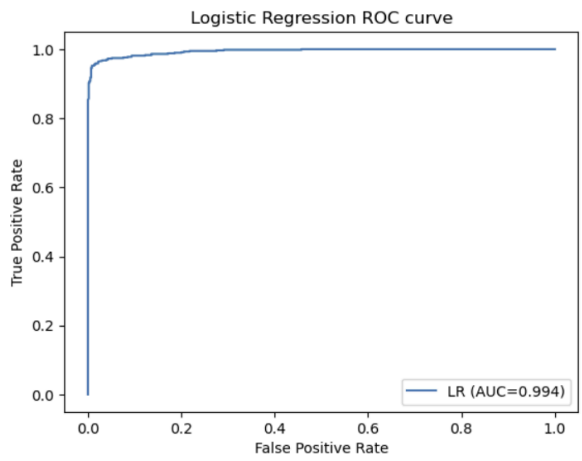


Figure 3:

a:



b:

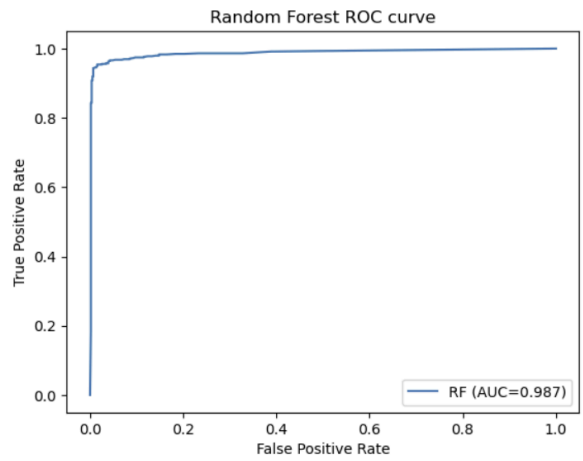
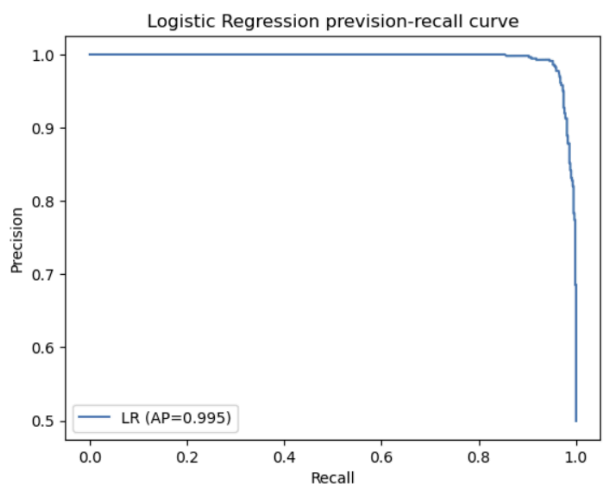


Figure 4:

a:



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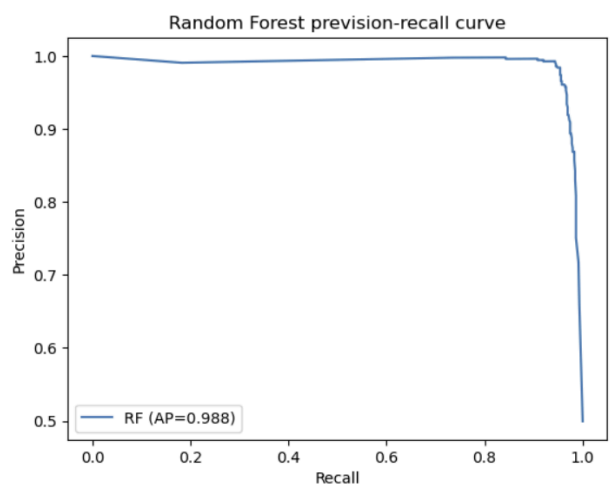
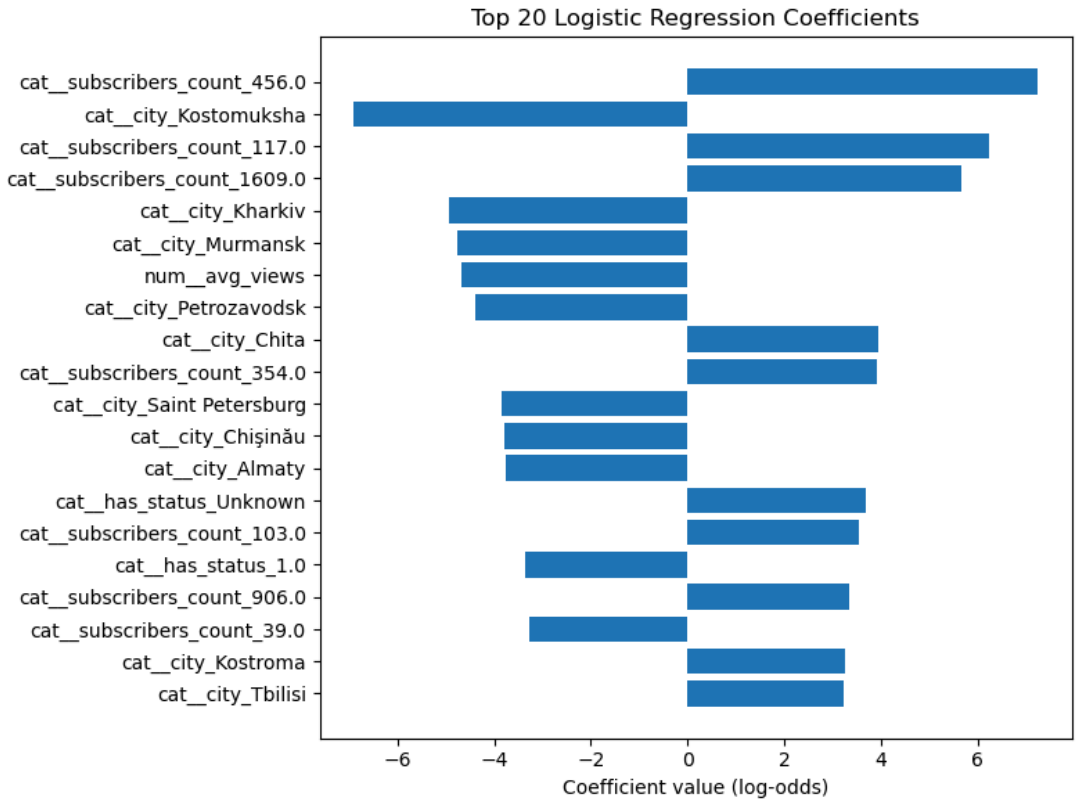


Figure 5:



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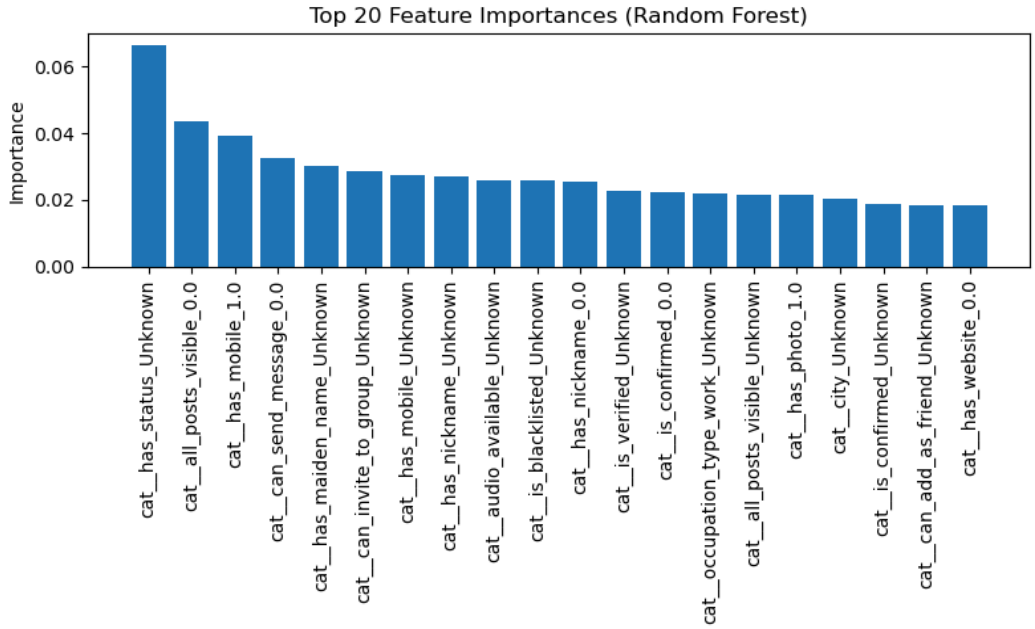

Figure 7:

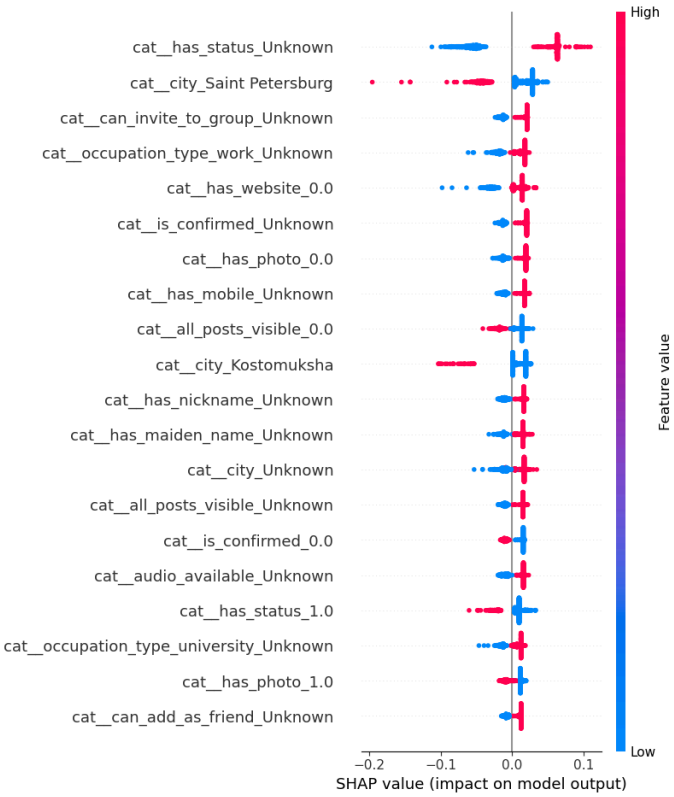

Figure 8:

Gender	Accuracy	Precision	Recall	F1-score	Support
Female(2.0)	0.953	0.960	0.954	0.957	278
Male(1.0)	0.975	0.993	0.956	0.973	888
Unknown	0.500	0.000	0.000	0.000	2
Other(0.0)	1.000	1.000	1.000	1.000	7